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| <h1>Standard Operating Procedure:<br/>SOP-Centrifuge-MethodA</h1> | <b>KENVUE OPERATIONAL STANDARD</b><br>Status: APPROVED<br>Version: 2.0 (Optimized) |
| Document ID: SOP-KNV-2026-OPT                                     | <b>Classification:</b> Confidential / Controlled                                   |

## 1.0 Purpose & Objective

The purpose of this Standard Operating Procedure (Standard Operating Procedure: SOP-Centrifuge-MethodA) is to define the standardized operational workflows, parameter verifications, and quality compliance criteria in accordance with Kenvue Manufacturing Quality Standards.

## 2.0 Scope, Applicability & Definitions

This procedure applies to all operational workstations, laboratory environments, and manufacturing personnel executing operations under Standard Operating Procedure: SOP-Centrifuge-MethodA. Standard definitions and approved terminologies are enforced across all operational lines.

## 3.0 Roles & Responsibilities

- Certified Operator: Execute procedural steps and verify operational parameters.
- Shift Supervisor: Oversee adherence to safety controls and approve deviations.
- Quality Assurance: Perform routine batch checks and review calibration records.

## 4.0 Mandatory Safety & PPE Requirements

### [MANDATORY SAFETY & PPE NOTICE]

All personnel must wear ANSI-certified safety glasses, nitrile gloves, and chemical-resistant protective gear. Do not bypass interlocks. Report any unexpected chemical exposure or mechanical anomalies immediately.

## 5.0 Equipment, Materials & Workstation Setup

Ensure all designated equipment, validated cleaning reagents, and lint-free wipes are staged in accordance with workstation staging guidelines prior to commencing operations.

## 6.0 Step-by-Step Operating Procedure

Standard Operating Procedure: High-Speed Centrifugation  
(Method A)

- 1.0 Purpose: Defines operating procedure for laboratory high-speed preparative centrifuge units.
- 2.0 Scope: Applies to all centrifugation runs in the active bio-pharmaceutical manufacturing facility.
- 3.0 Responsibilities: Operators must be certified. Supervisors must verify equipment calibration log.

- 4.0 Equipment & Materials: Model Beckman J2-21 Centrifuge, Fixed-angle JA-10 rotor, Balanced sample bottles.
- 5.0 Pre-Operational Checks: Visually inspect rotor for cracks or corrosion. Verify rubber O-ring seal integrity.
- 6.0 Procedure:
- 6.1 Balance opposing centrifuge bottles with analytical precision within  $\pm 0.05$  grams.
- 6.2 Secure rotor onto drive spindle with locking knob. Ensure seated flush without wobble.
- 6.3 Set centrifugation speed to 4500 RPM. Set chamber cooling temperature to 4°C.
- 6.4 Set run duration to 25 minutes. Engage vacuum latch mechanism.
- 6.5 Press Start. Observe acceleration curve until target speed is achieved.
- 6.6 Upon cycle completion and deceleration to 0 RPM, vent vacuum chamber and carefully retrieve bottles.
- 7.0 Documentation: Record run date, speed (4500 RPM), operator initials, and lot ID in equipment log book.

#### Operating Guide: High-Speed Centrifugation Protocol (Method B)

- 1.0 Purpose and Objective: Standard guidelines for executing high-speed centrifuge separation processes.
- 2.0 Scope: Applicable to all production and QC laboratory centrifuge equipment.
- 3.0 Operating Personnel: Only trained production associates may operate centrifuge machinery.
- 4.0 Apparatus: High-speed preparative centrifuge, JA-10 rotor assembly, balance scale.
- 5.0 Inspection Steps: Inspect the rotor well for debris. Inspect rotor O-ring and lubricate lightly if dry.
- 6.0 Operational Procedure:
- 6.1 Accurately balance sample containers against counterpart bottles on twin-pan balance ( $\pm 0.05$  g).
- 6.2 Mount rotor onto drive hub and tighten retaining nut firmly clockwise.
- 6.3 Program rotational velocity to 5000 RPM. Set refrigeration system to 2°C.
- 6.4 Program timer for 30 minutes. Secure vacuum barrier door.
- 6.5 Initiate rotation. Monitor instrument sound and vibration levels during ramp-up.
- 6.6 Following complete deceleration, disengage safety lock and extract sample containers.
- 7.0 Batch Record Entry: Document operator signature, actual rotational speed (5000 RPM), and temperature in batch log.

#### Standard Operating Procedure: Secondary Blister Packaging & Cartoning

- 1.0 Purpose: Guidelines for packa

## 7.0 Quality Control, Inspection & Acceptance Standards

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Verify that all operating parameters and surface cleanliness checks remain strictly within quantified acceptance criteria (100% compliance required before release).

## 8.0 Documentation, Records Retention & References

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All execution records, deviations, and supervisory sign-offs must be logged in the batch documentation system and retained per corporate document retention schedules.

## 9.0 Revision History & Document Control

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Document Title: Standard Operating Procedure: SOP-Centrifuge-MethodA  
Document ID: SOP-KNV-2026-OPT

Version: 2.0 (Optimized)

Status: Approved for Operational Implementation.

Change Summary: Restructured to 10-tier Kenvue standard architecture, active imperative commands, and validated quality tolerances.

## 10.0 Appendix: Operational Checklists & Logsheets

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### **[MANDATORY SAFETY & PPE NOTICE]**

The following operational checklist and completion logsheet forms must be executed and retained in the official batch record upon procedure completion.

## 10.1 Custom Operational Directives

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Mandatory User Requirements:

Add double sign-off on tare balance weight.